Quantification: 682_z29
=======================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	12
interior lacuna count	10
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 10)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	1885	1629	856.283	px^2
major_axis_length_px	lacuna shape	79.6556	78.7239	19.8072	px
minor_axis_length_px	lacuna shape	30.5623	31.1018	8.0704	px
aspect_ratio	lacuna shape	2.7208	2.6218	0.7334	unitless
eccentricity	lacuna shape	0.9094	0.9241	0.0653	unitless
solidity	lacuna shape	0.8664	0.8806	0.0554	unitless
roots_count	headline	8.9	9	3.8137	count
ring_length_r30_px	headline	337	327.5	121.485	px
ring_length_r60_px	reported	829.4	816.5	309.239	px
owned_length_px	ownership-dependent	2597.01	2350.15	2113.33	px
edge_count	ownership-dependent	113.7	94.5	101.02	count
mean_edge_length_px	ownership-dependent	24.0819	23.6309	3.8213	px
perimeter_px	normalised	214.094	221.823	49.7159	px
ring_area_r30_px2	normalised	8566.8	8616	1406.48	px^2
ring_area_r60_px2	normalised	21496.2	20989	3247.7	px^2
in_frame_fraction_r30	normalised	0.9938	1	0.0197	unitless
in_frame_fraction_r60	normalised	0.9805	1	0.0615	unitless
ring_density_r30	normalised	0.0386673	0.0389188	0.0104498	px^-1
ring_density_r60	normalised	0.0376959	0.0375964	0.0097994	px^-1
roots_per_100px_perimeter	normalised	4.1284	3.7069	1.5491	per 100 px
ring_attached_length_r30_px	network v2	308.1	310	124.339	px
ring_attached_length_r60_px	network v2	675.3	641.5	335.048	px
ring_length_w_r30_px	network v2	378.603	371.614	132.981	px
ring_length_w_r60_px	network v2	934.567	916.746	347.651	px
sholl_crossings_r10	network v2	8.2	8	2.8597	count
sholl_crossings_r20	network v2	10.7	11	3.6225	count
sholl_crossings_r30	network v2	12	11	3.7417	count
loop_count	loops, report only	12.8	6.5	17.3897	count
loop_edge_share	loops, report only	0.2591	0.2583	0.2207	share
ring_width_mean_r30_px	width	7.1512	7.1595	0.479	px

Per field
----

measure	value	units
analysed_area_px2	1.02386e+06	px^2
total_skeleton_length_px	35078	px
canalicular_length_density_per_px	0.0342605	px^-1
junction_count	759	count
junction_density_per_px2	0.00074131	px^-2
lacuna_count	12	count

lacunae_per_px2	1.172e-05	px^-2
skeleton_component_count	232	count
mean_component_length_px	151.198	px
median_component_length_px	21	px
field_length_density_w_per_px	0.0387192	px^-1
field_density_without_flagged_per_px	0.0352474	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	513	px
unattached_component_count	32	count
canalicular_length_density_attached_per_px	0.0342605	px^-1
field_length_density_with_unattached_per_px	0.0347615	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	151	count
loop_edge_share	0.3353	share
edge_count	1694	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6694	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10	px
width_pixels_used	33421	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2083	86.508	34.896	2.479	0.915	0.8266	1.5435	13.4983	99.4023	True
2	2213	89.1614	32.1486	2.7734	0.9327	0.8778	-1.1291	134.914	921.292	False
3	3140	120.617	34.4194	3.5043	0.9584	0.8885	1.5609	431.048	220.093	False
4	2837	88.9183	41.496	2.1428	0.8844	0.9229	-1.5453	532.043	493.055	False
5	1634	75.9278	30.0551	2.5263	0.9183	0.8211	-1.4797	555.483	357.187	False
6	3781	121.501	43.6088	2.7862	0.9334	0.8771	1.4272	590.482	55.0987	True
7	1399	81.52	23.1918	3.515	0.9587	0.8541	1.5542	716.135	181.57	False
8	663	61.7115	15.8655	3.8897	0.9664	0.7408	-1.1603	728.596	861.06	False
9	1130	64.3164	23.6688	2.7173	0.9298	0.8835	1.4062	842.292	125.237	False
10	1244	49.7173	33.3792	1.4895	0.7411	0.9325	-1.542	904.6	760.807	False
11	2966	92.1224	41.6334	2.2127	0.8921	0.895	-1.4899	920.401	307.385	False
12	1624	72.5441	29.7651	2.4372	0.9119	0.8476	-0.945	978.497	437.615	False

Network per lacuna

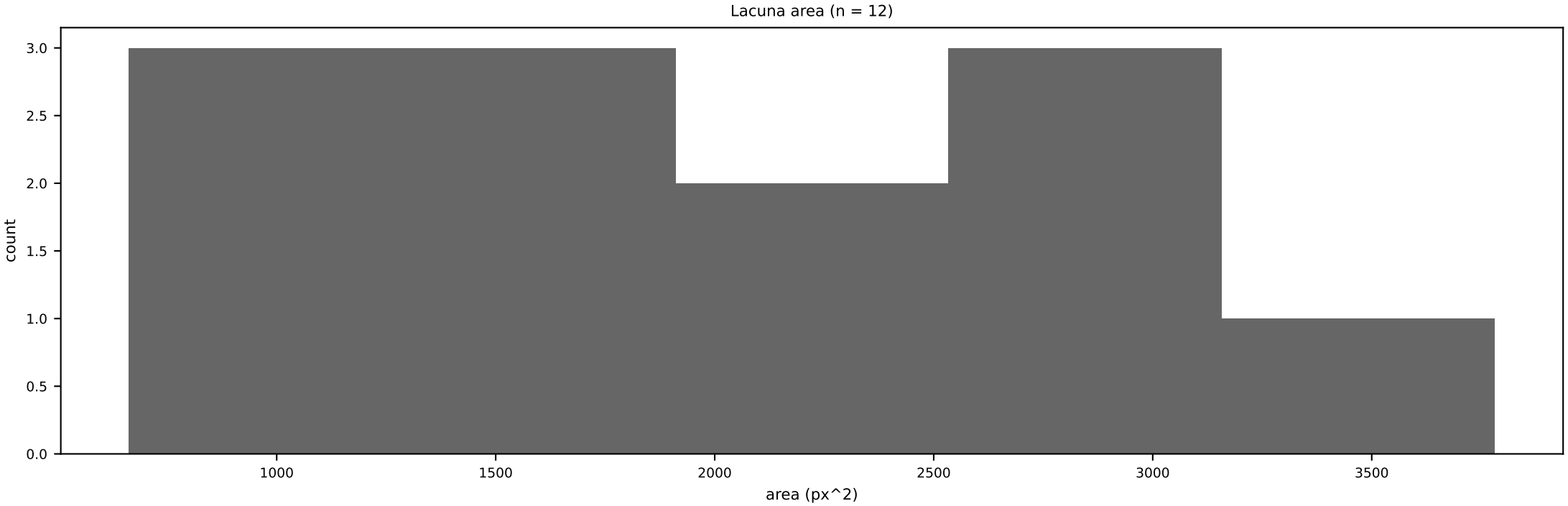
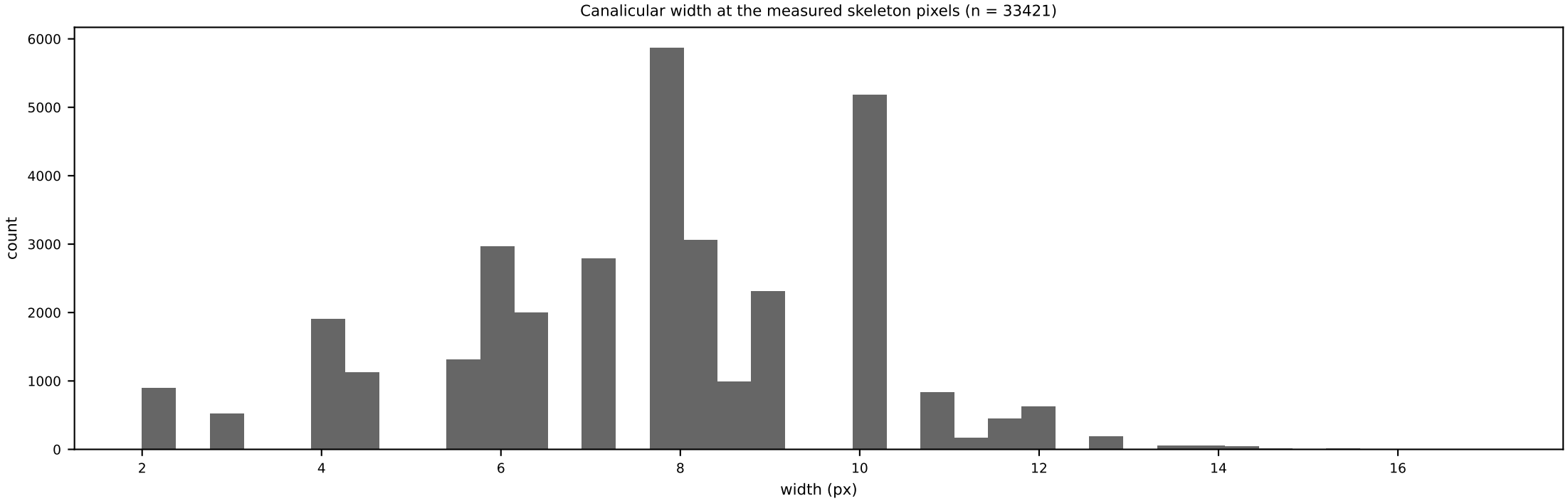
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	192	359	1495.72	79	18.9332	
2	17	536	1315	6300.68	276	22.8286	7.2892
3	12	476	1334	5907.6	299	19.7579	7.7077
4	7	268	652	3308.06	134	24.687	7.1937
5	9	432	911	2707.02	120	22.5585	7.3207

6	11	392	836	1426.67	64	22.2918	7.4798
7	8	376	896	1993.28	73	27.3052	6.2524
8	9	214	646	2834.25	116	24.4332	6.9231
9	5	279	776	507.446	16	31.7154	7.9745
10	3	176	516	645.47	34	18.9844	6.8366
11	9	385	857	1397.99	52	26.8844	6.8891
12	10	228	391	368.291	17	21.6642	7.1254

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	262.096	5607	13819	0.5809	0.5584	0.0342429	0.0259787	1.5262
2	229.522	9151	23871	1	0.999	0.0585728	0.0550878	7.4067
3	309.279	11331	28206	1	1	0.0420087	0.0472949	3.88
4	235.238	9387	21888	1	1	0.0285501	0.029788	2.9757
5	214.167	8637	20090	1	1	0.0500174	0.0453459	4.2023
6	322.836	9940	22198	0.8402	0.7724	0.0394366	0.0376611	3.4073
7	229.48	8595	22054	1	1	0.0437464	0.0406275	3.4861
8	156.853	6922	19429	1	1	0.0309159	0.0332493	5.7379
9	161.817	7292	19909	1	1	0.0382611	0.0389774	3.0899
10	147.196	6914	19480	1	1	0.0254556	0.0264887	2.0381
11	254.693	9728	23664	1	1	0.0395765	0.0362154	3.5337
12	202.693	7711	16371	0.9376	0.8055	0.0295682	0.0238837	4.9336

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	85	150	219.51	395.664	2	6	6
2	528	1263	590.333	1486.51	12	17	18
3	450	1208	539.304	1503.74	13	14	16
4	257	512	292.51	715.931	6	10	11
5	361	719	481.534	1022.01	9	11	15
6	379	771	440.563	939.968	10	12	12
7	358	644	417.664	999.281	10	12	10
8	191	482	252.037	750.382	5	7	9
9	278	721	325.563	873.654	7	11	11
10	127	316	199.995	597.085	5	7	8
11	342	639	432.363	959.838	9	13	15
12	189	249	254.723	437.233	6	5	7

682_z29: distributions, pre-validation, px